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Hence, the amount of energy required for the formation of 1 mole of methane is -74.7 kJ

The heat of formation methane = -74.7 kJ mol^{-1}

(2) Calculation of calorific value of food and fuels: The calorific value is defined as the amount of heat produced in calories (or joules) when one gram of the substance is completely burnt. The SI unit of calorific value is J kg^{-1}. However, it is usually expressed in cal g^{-1}.

Heat of solution:

Heat changes are usually observed when a substance is dissolved in a solvent. The heat of solution is defined as the change in enthalpy when one mole of a substance is dissolved in a specified quantity of solvent at a given temperature.

Heat of neutralisation:

The heat of neutralisation is defined as "The change in enthalpy when one gram equivalent of an acid is completely neutralised by one gram equivalent of a base or vice versa in dilute solution".

$HCl(aq)+NaOH(aq) \rightarrow NaCl\,(aq)+ H_2O(l)$

$\Delta H = -57.32$ kJ

$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$

$\Delta H = -57.32$ kJ

The heat of neutralisation of a strong acid and strong base is around – 57.32 kJ, irrespective of nature of acid or base used which is evident from the below mentioned examples.

$HCl\,(aq) + KOH(aq) \rightarrow KCl\,(aq) + H_2O(l)$

$\Delta H = -57.32$ kJ

$HNO_3(aq)+KOH(aq) \rightarrow KNO_3(aq)+ H_2O(l)$

$\Delta H = -57.32$ kJ

$H_2SO_4(aq) + 2KOH(aq) \rightarrow$

$K_2SO_4(aq) + 2\,H_2O(l)$

$\Delta H = -57.32$ kJ

The reason for this can be explained on the basis of Arrhenius theory of acids and bases which states that strong acids and strong bases completely ionise in aqueous solution to produce H^+ and OH^- ions respectively. Therefore in all the above mentioned reactions the neutralisation can be expressed as follows.

$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$

$\Delta H = -57.32$ kJ

Molar heat of fusion

The molar heat of fusion is defined as "the change in enthalpy when one mole of a solid substance is converted into the liquid state at its melting point".

For example, the heat of fusion of ice can be represented as

$H_2O(s) \xrightarrow{273K} H_2O(l)\ \Delta H_{fusion} = +5.98$kJ

Molar heat of vapourisation

The molar heat of vaporisation is defined as "the change in enthalpy when